

Inverse Problems in Geophysics

Exercise 2: Least-squares

2. MGPY+MGIN

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Task from last time

$$m_1 - m_2 = -1$$

$$2m_1 - m_2 = 0$$

$$m_1 + m_2 = 2.5$$

1. Set up the matrix $\mathbf{G}$ and the data vector $\mathbf{d}$
2. Reformulate equations into $m_2(m_1)$
3. Draw the three equations for a range
4. Guess the “best tradeoff”
5. Add error ($d + \delta d$) and draw shaded region
6. Increase the error until the regions meet (how big is the common error?)
7. Use a range of values for m_1 & m_2 , show the objective function and detect minimum

The objective function

We minimize the L2-norm (summed squared distances) of the residual between data and model response:

$$\Phi = \|\mathbf{d} - \mathbf{G}\mathbf{m}\|_2^2 = \sum_1^N (\mathbf{d}_i - \mathbf{g}_i(\mathbf{m}))^2 \rightarrow \min$$



Grid search

Computing the objective function for value ranges and detecting the minimum.

Resolution matrix

1. Compute the least squares inverse matrix

$$\mathbf{G}_{LS}^{\dagger} = (\mathbf{G}^T \mathbf{G})^{-1} \mathbf{G}^T$$

2. Compute the least-squares solution
3. Display the solution in the m_1/m_2 cross-plot with optimum errors
4. Mark the solution in the objective function plot
5. Compute the formal model resolution matrix $\mathbf{R}^M$ and the data information matrix $\mathbf{R}^D$
6. Display both matrices using a meaningful color scale

Linear regression

Consider the problem of linear regression $y = a + bx$

What's the model? What's the data & model response?

How can we formulate this as a matrix problem ($\mathbf{G}\mathbf{m} = \mathbf{d}^{syn}$)?

Tasks

1. Set up a synthetic model, create data and add noise
2. Compute the least-squares solution
3. Plot the model response along with the data
4. Determine the misfit
5. Compute the data information matrix and plot it
6. Plot the data importance (diagonal) over x

Problem types

Example linear problem: Traveltime tomography

$$t = \int_L \frac{1}{v} dl = \int_L s dl$$

piece-wise constant s : $t = \sum_j s_j \Delta l_j \Rightarrow t_i = \sum W_{ij} s_j$

$\mathbf{d} = \mathbf{W}\mathbf{m}$ where $\mathbf{W}$ contains lengths of rays (rows) in cells (columns)

with $\mathbf{d} = [t_1, \dots, t_N]$, $\mathbf{m} = [s_1, \dots, s_M]$

Types of inverse problems

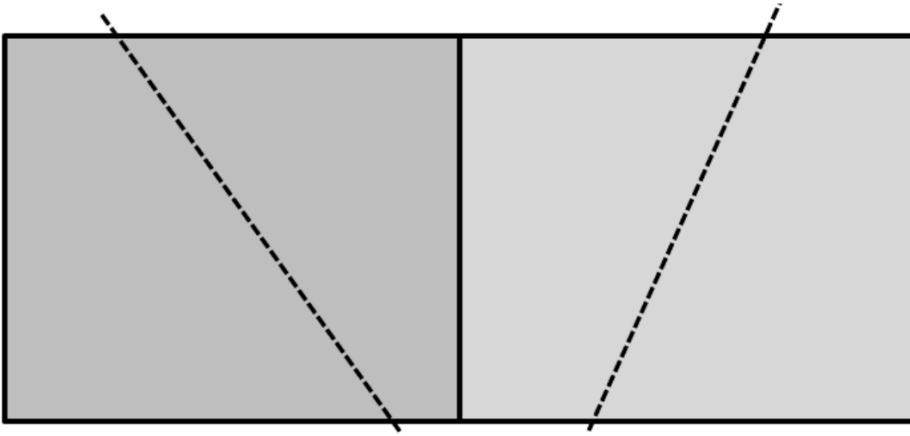
$$\textcircled{!} \mathbf{d} = \mathbf{G}\mathbf{m} + \mathbf{n} \Rightarrow \mathbf{G} \in \mathcal{R}^{N \times M}$$

rows stand for measurements, columns represent model parameters

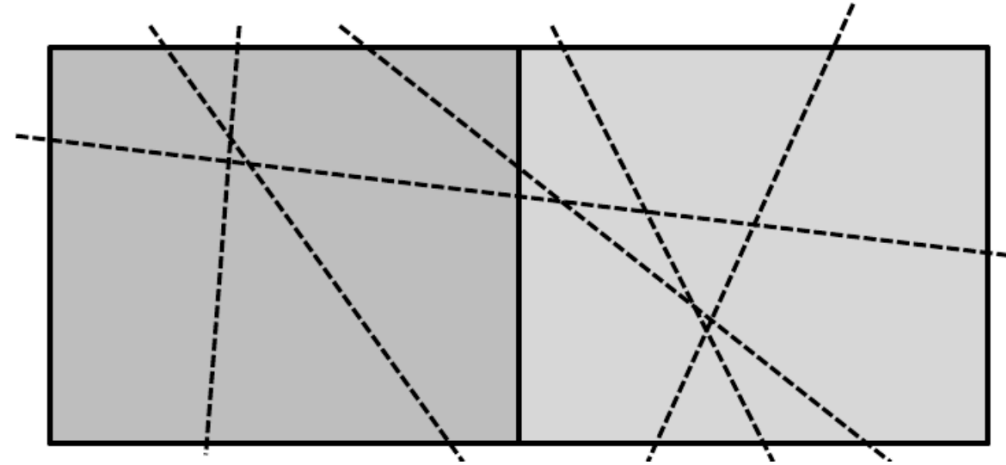
- $N = M$: even-determined
- $N > M$: over-determined problem (no existing solution)
- $N < M$: under-determined problem (no unique solution)
- mostly: (both over- and) under-determined model parts

Types of inverse problems (Menke, 2012)

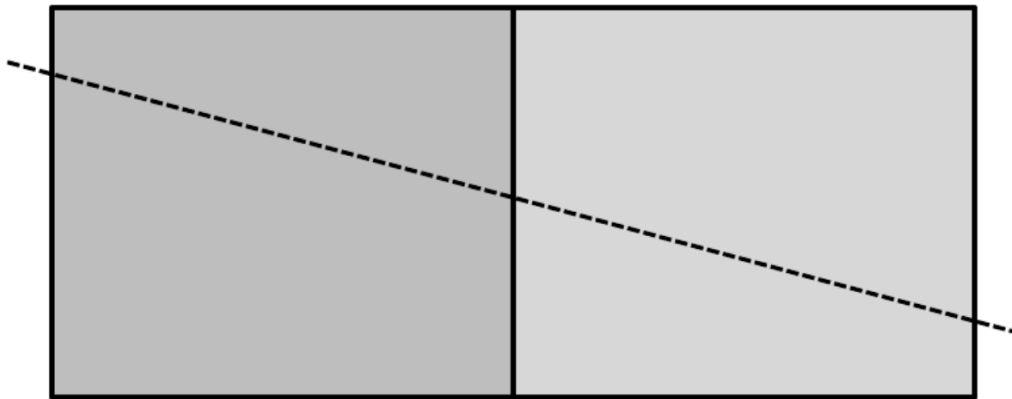
Even-determined



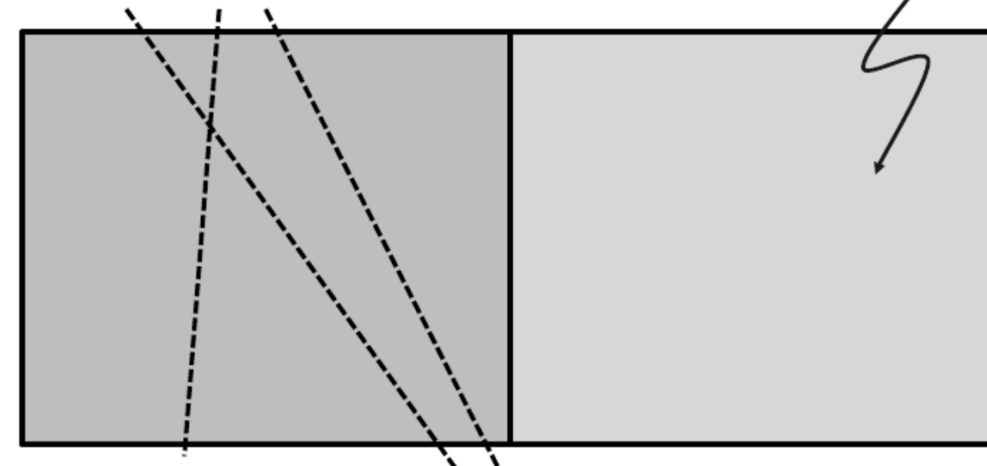
Over-determined



Under-determined



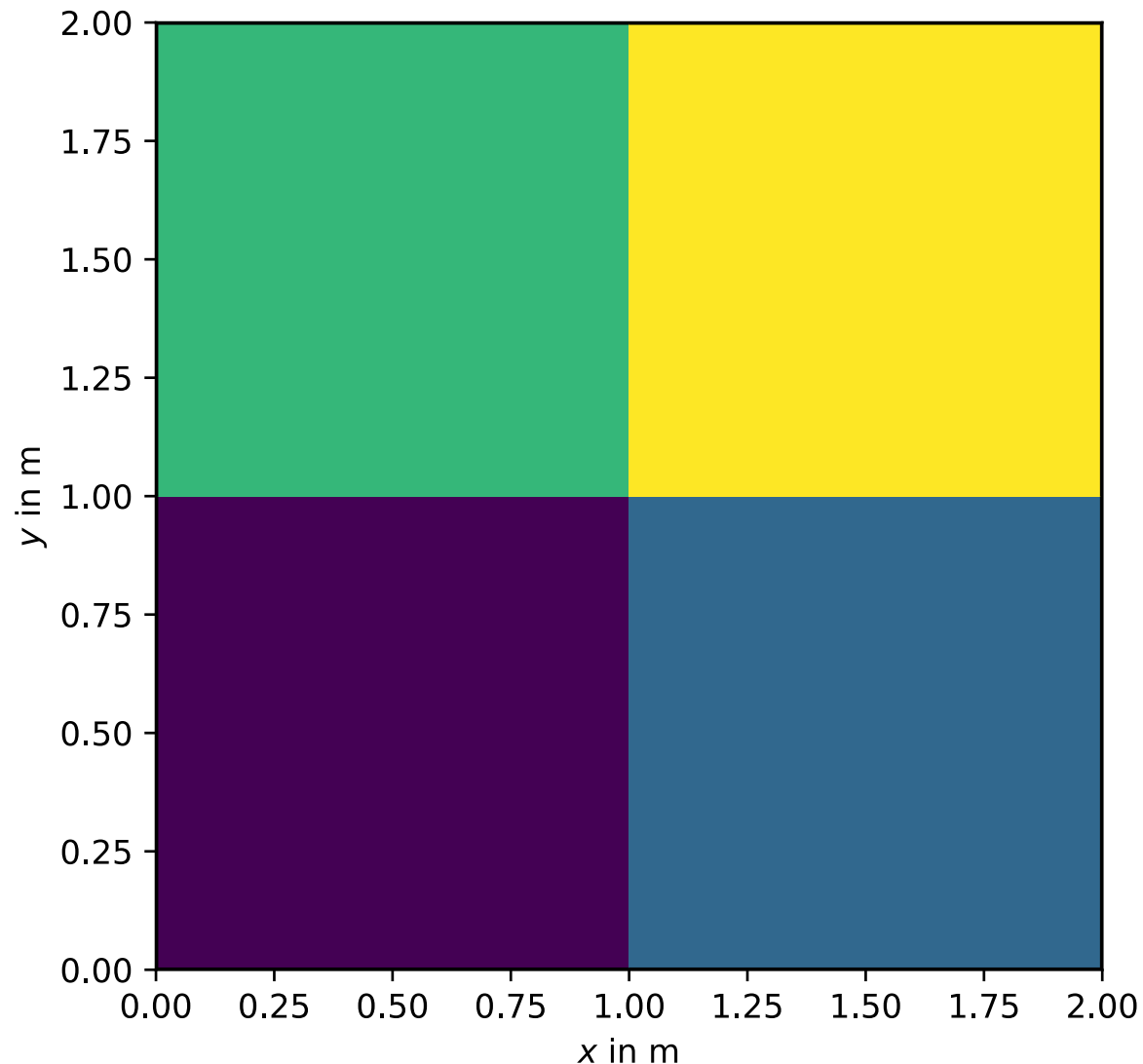
Mixed-determined



null space

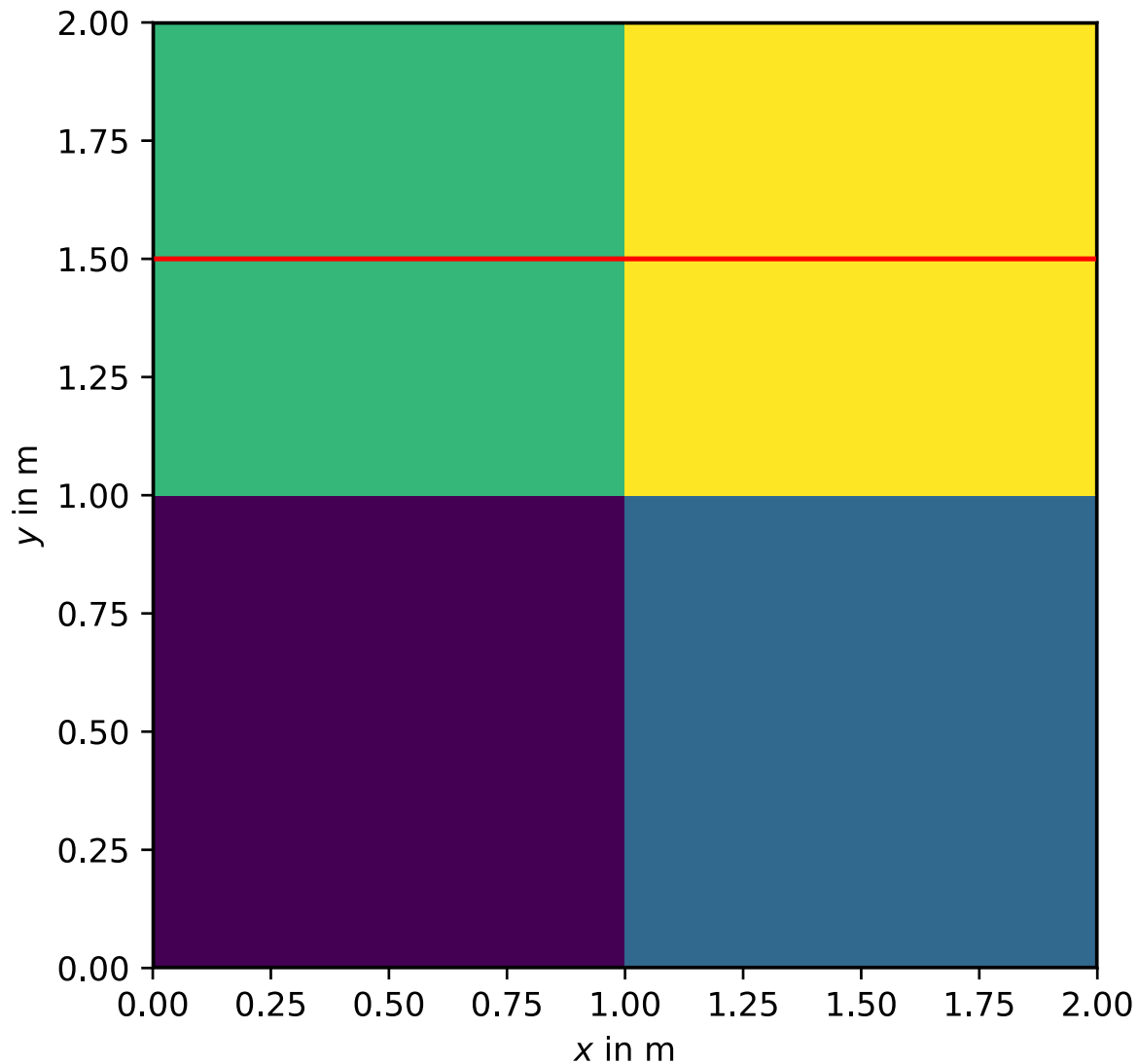


Problem types - model



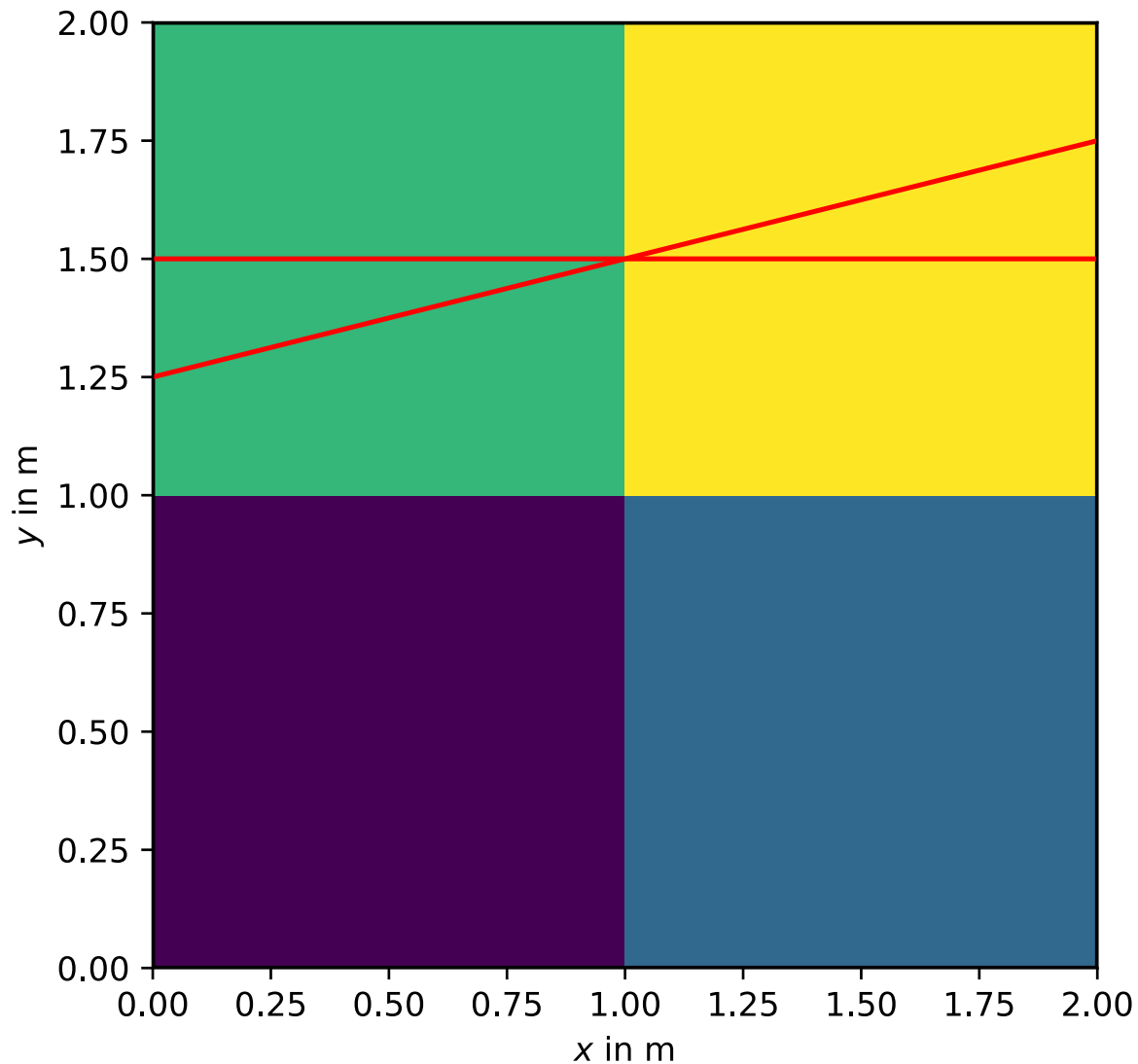
- guess whether it is a) over-, b) under-, or c) mixed-determined
- generate the ray path matrix
- determine the rank by using SVD
- generate data from an inhomogeneous model vector
- choose method & compute model
- compare with the backslash
- compile the numbers in a table

Problem types - 1



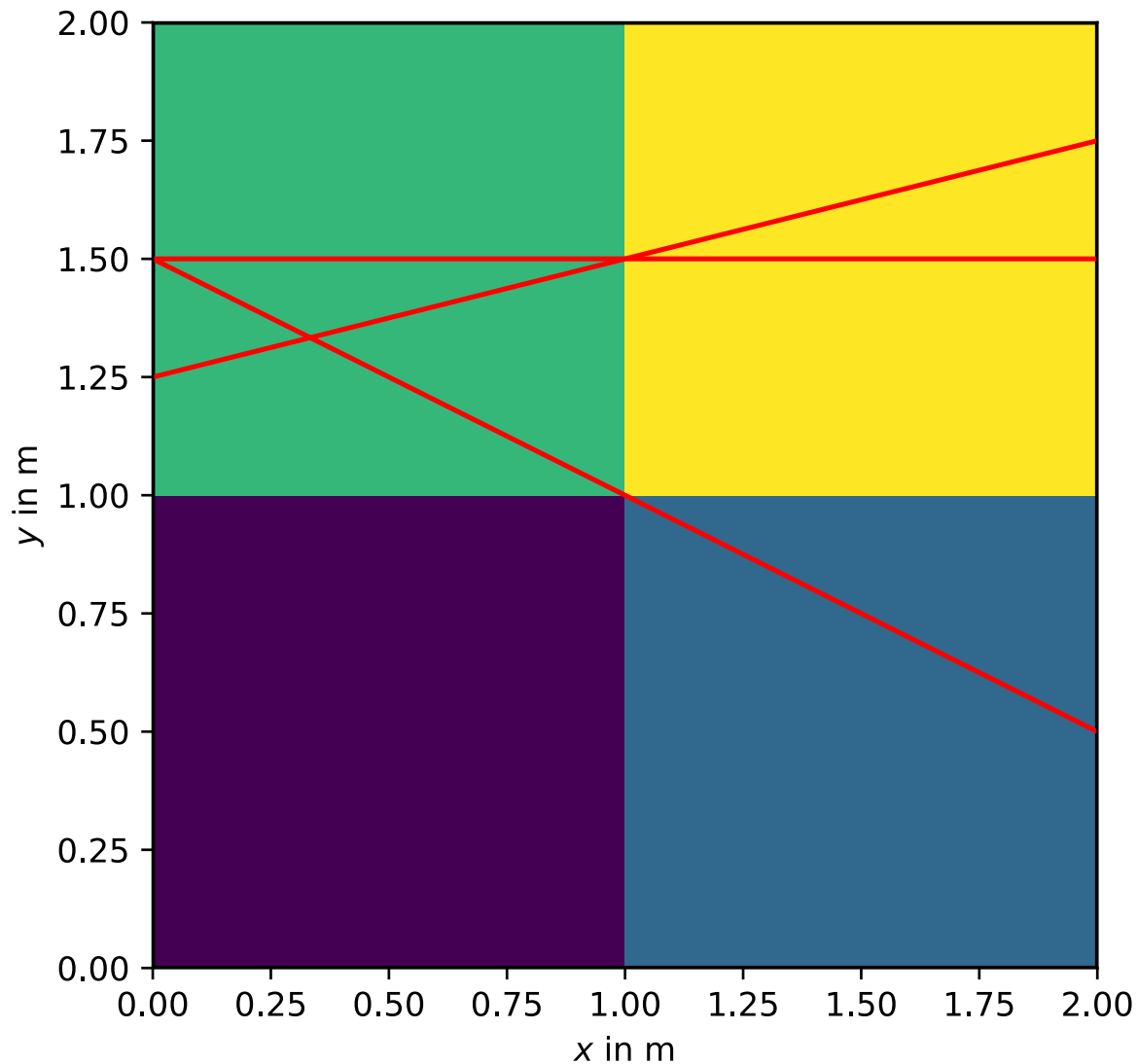
- guess whether it is a) over-, b) under-, or c) mixed-determined
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Problem types - 2



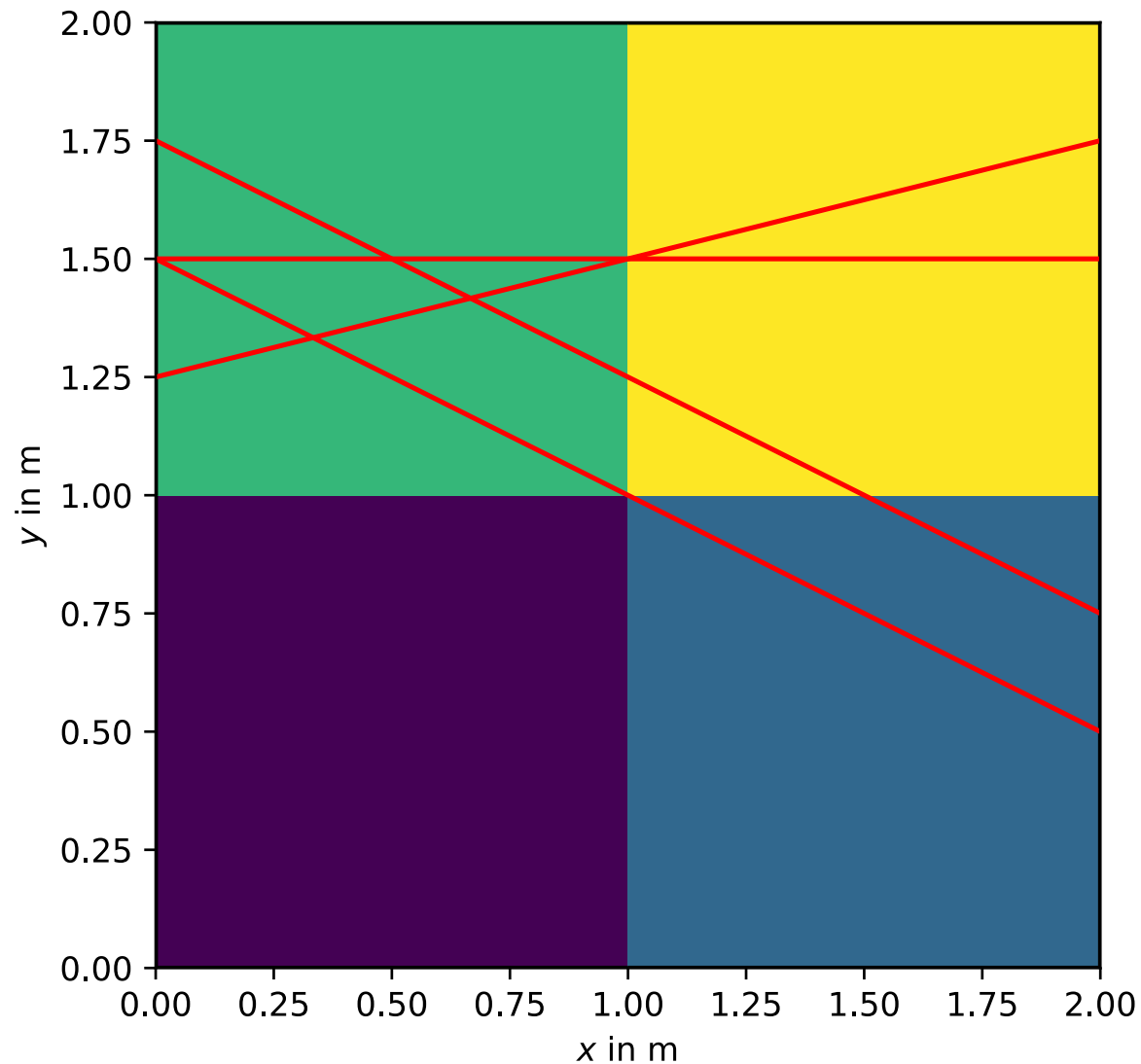
- guess whether it is a) over-, b) under-, or c) mixed-determined
- generate the ray path matrix
- determine the rank by using SVD
- generate data from an inhomogeneous model vector
- choose method & compute model
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Problem types - 3



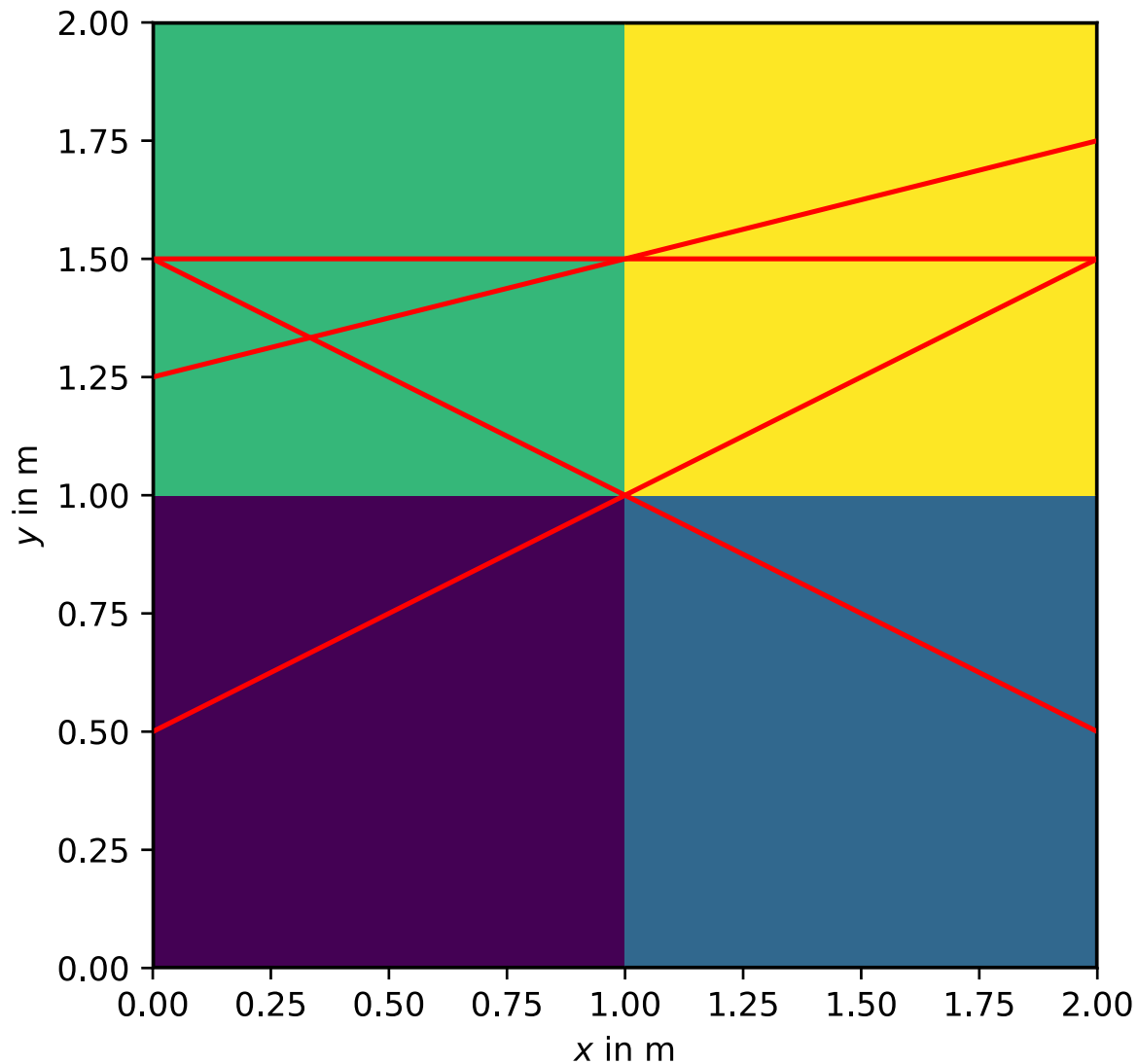
- guess whether it is a) over-, b) under-, or c) mixed-determined
- generate the ray path matrix
- determine the rank by using SVD
- generate data from an inhomogeneous model vector
- choose method & compute model
- compare with the backslash
- compile the numbers in a table

Problem types - 4



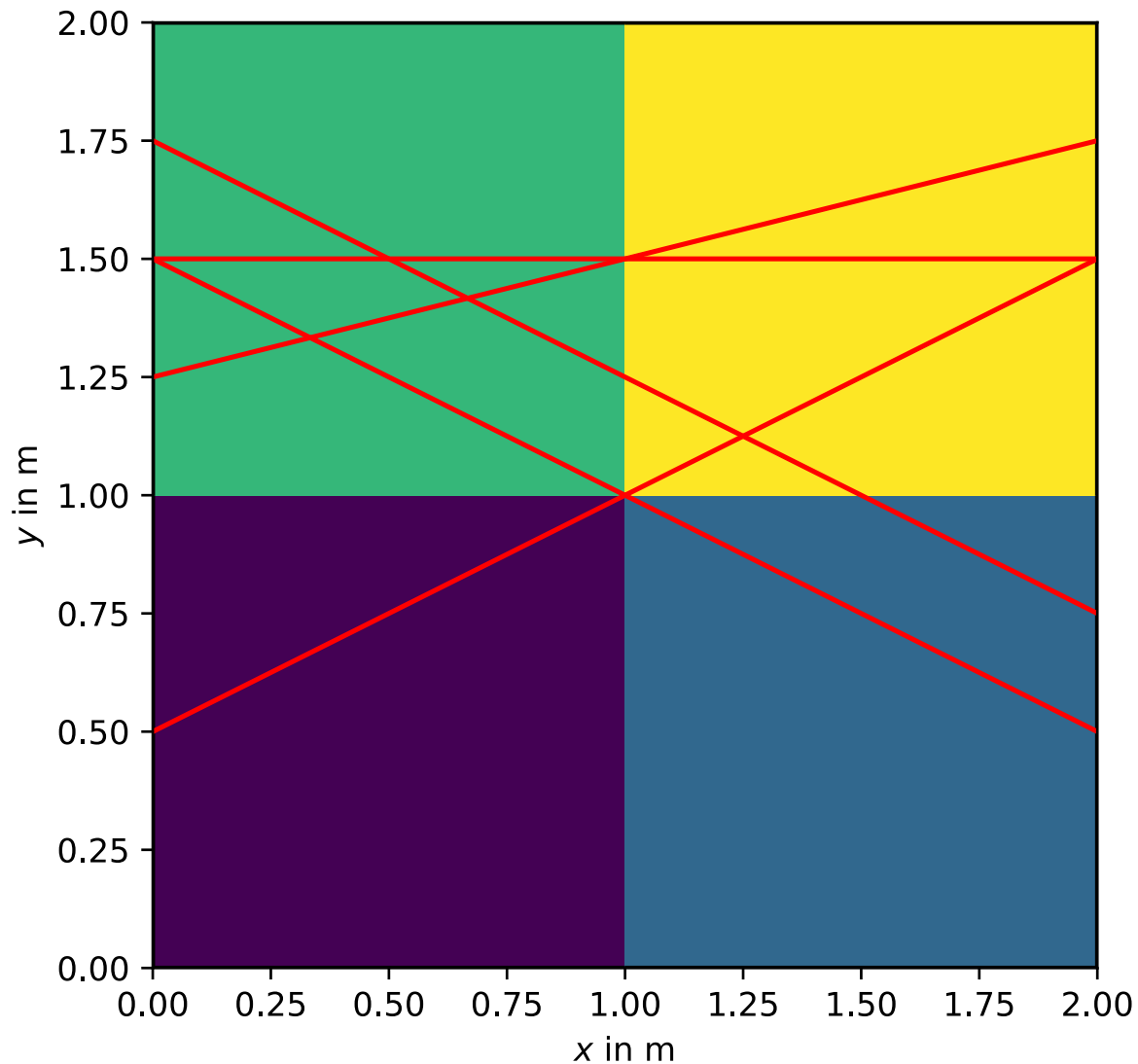
- guess whether it is a) over-, b) under-, or c) mixed-determined
- generate the ray path matrix
- determine the rank by using SVD
- generate data from an inhomogeneous model vector
- choose method & compute model
- compare with the backslash
- compile the numbers in a table

Problem types - 5



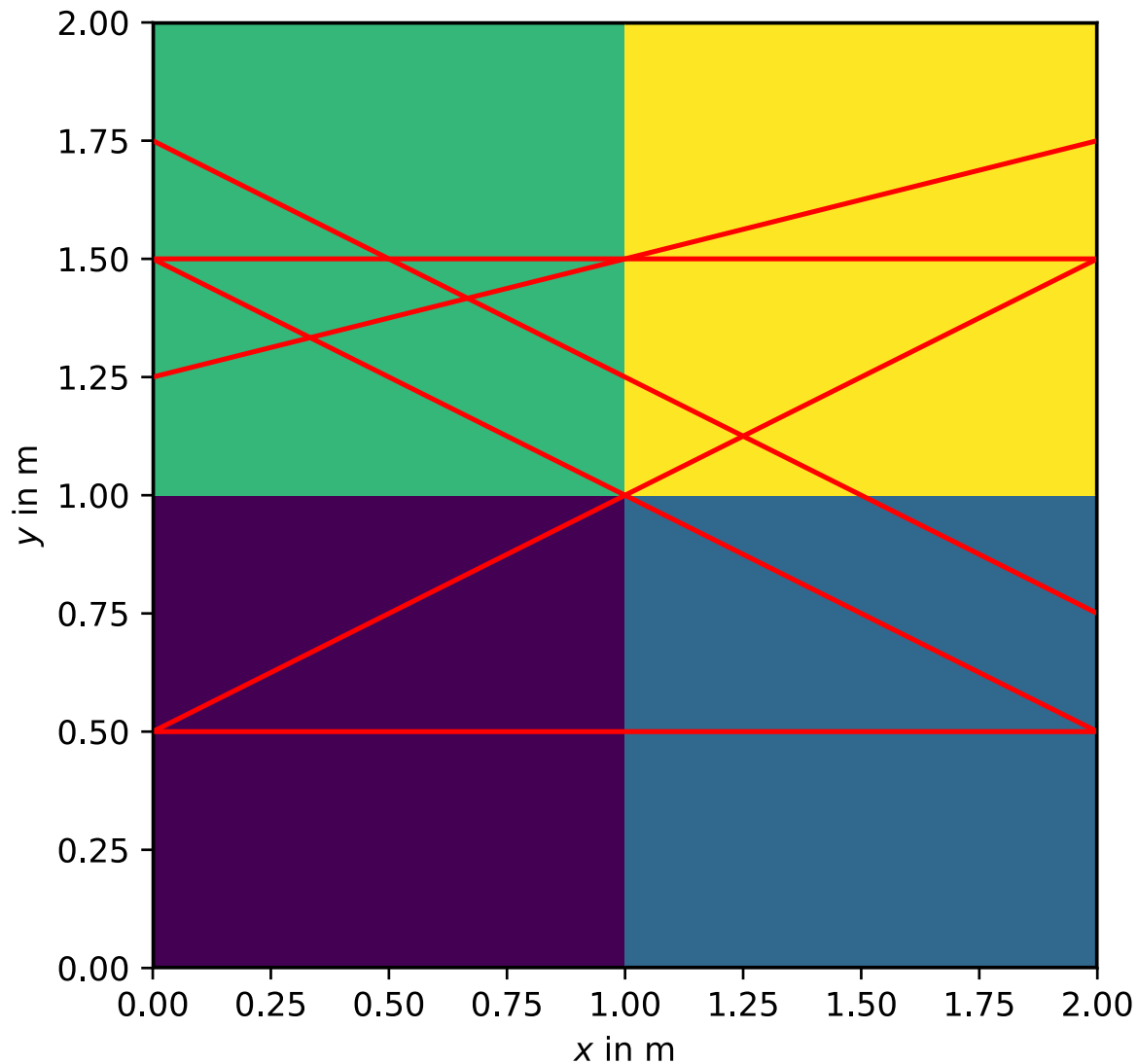
- guess whether it is a) over-, b) under-, or c) mixed-determined
- generate the ray path matrix
- determine the rank by using SVD
- generate data from an inhomogeneous model vector
- choose method & compute model
- compare with the backslash
- compile the numbers in a table

Problem types - 6



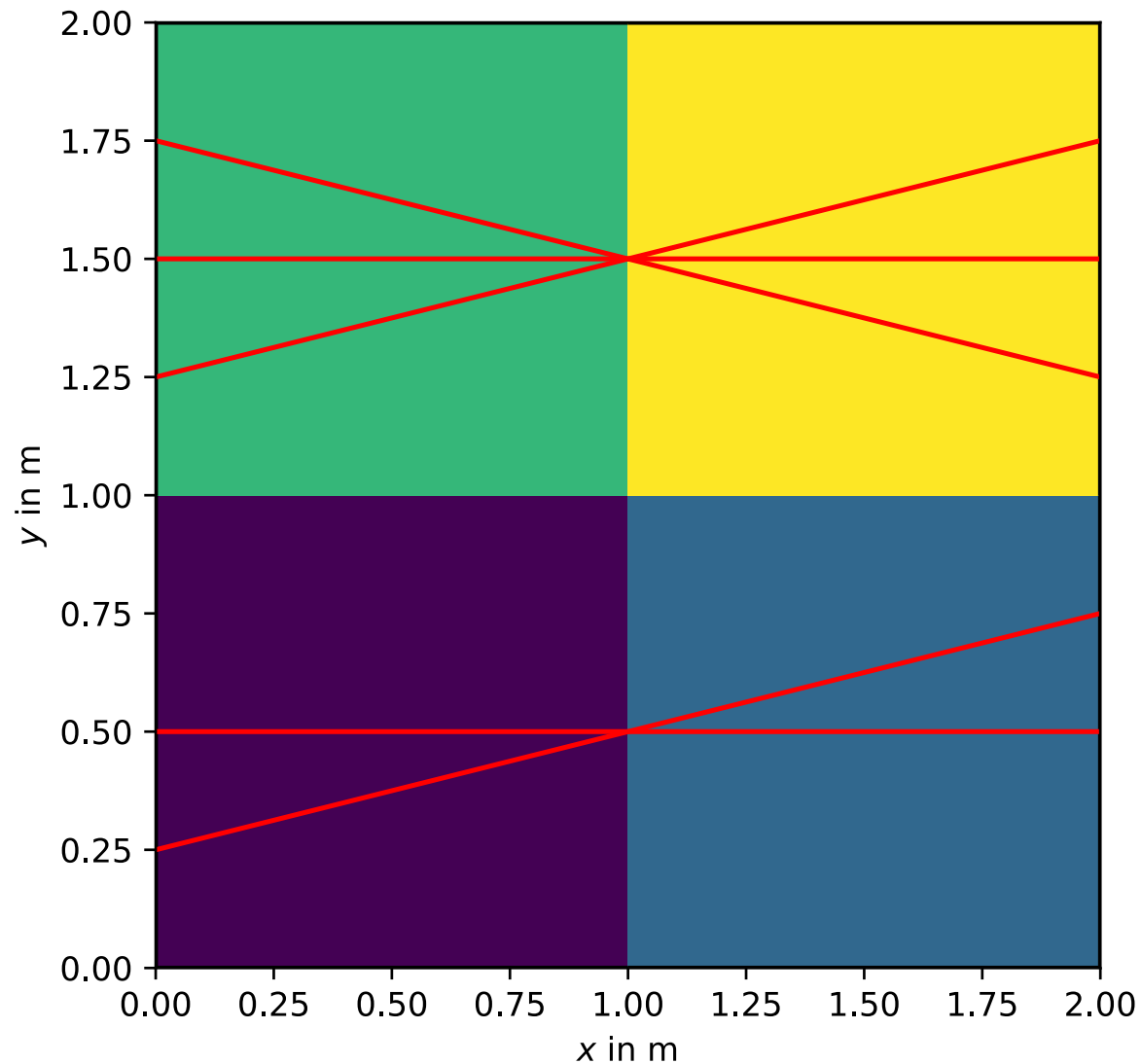
- guess whether it is a) over-, b) under-, or c) mixed-determined
- generate the ray path matrix
- determine the rank by using SVD
- generate data from an inhomogeneous model vector
- choose method & compute model
- compare with the backslash
- compile the numbers in a table

Problem types - 7



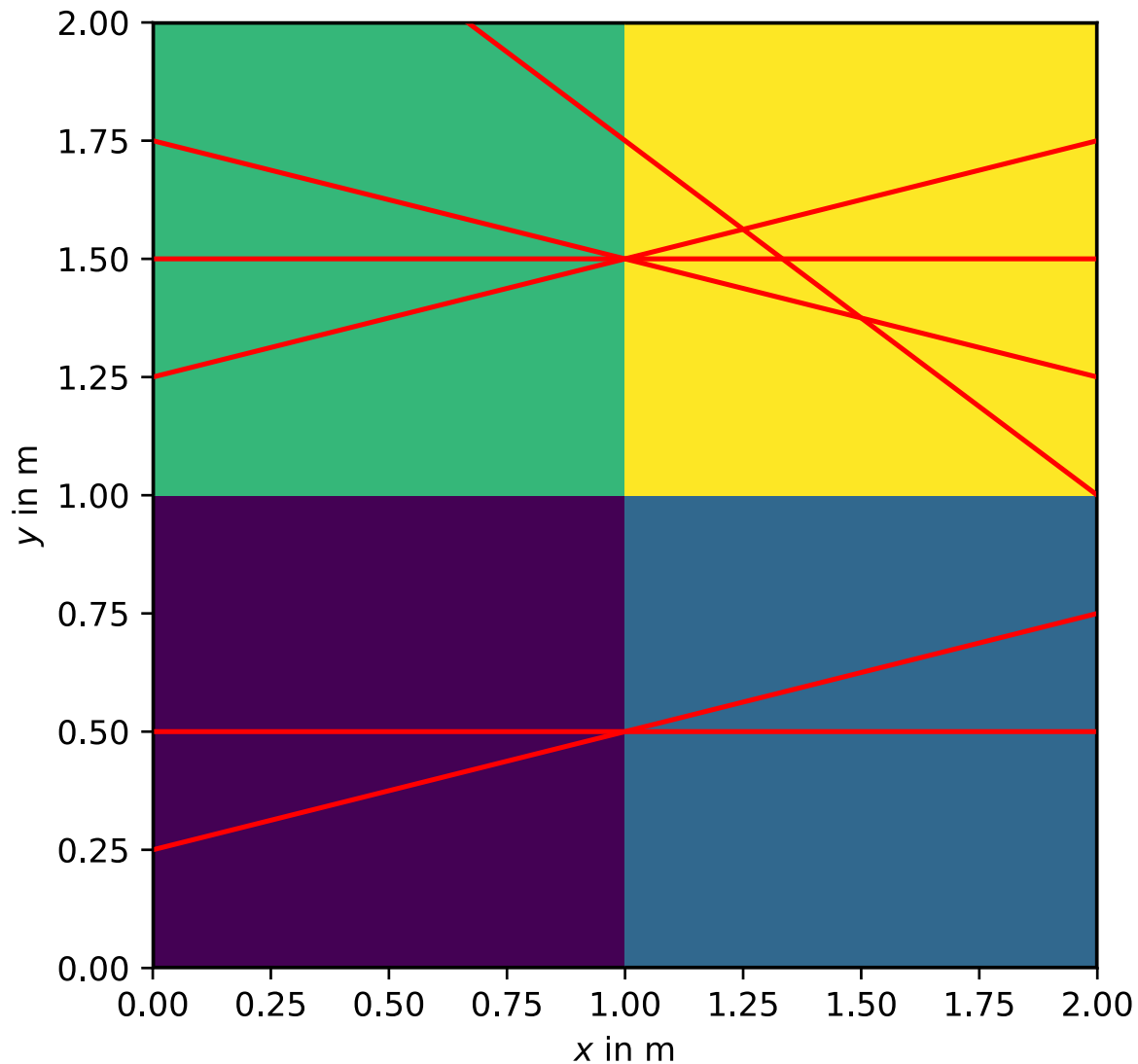
- guess whether it is a) over-, b) under-, or c) mixed-determined
- generate the ray path matrix
- determine the rank by using SVD
- generate data from an inhomogeneous model vector
- choose method & compute model
- compare with the backslash
- compile the numbers in a table

Problem types - 8



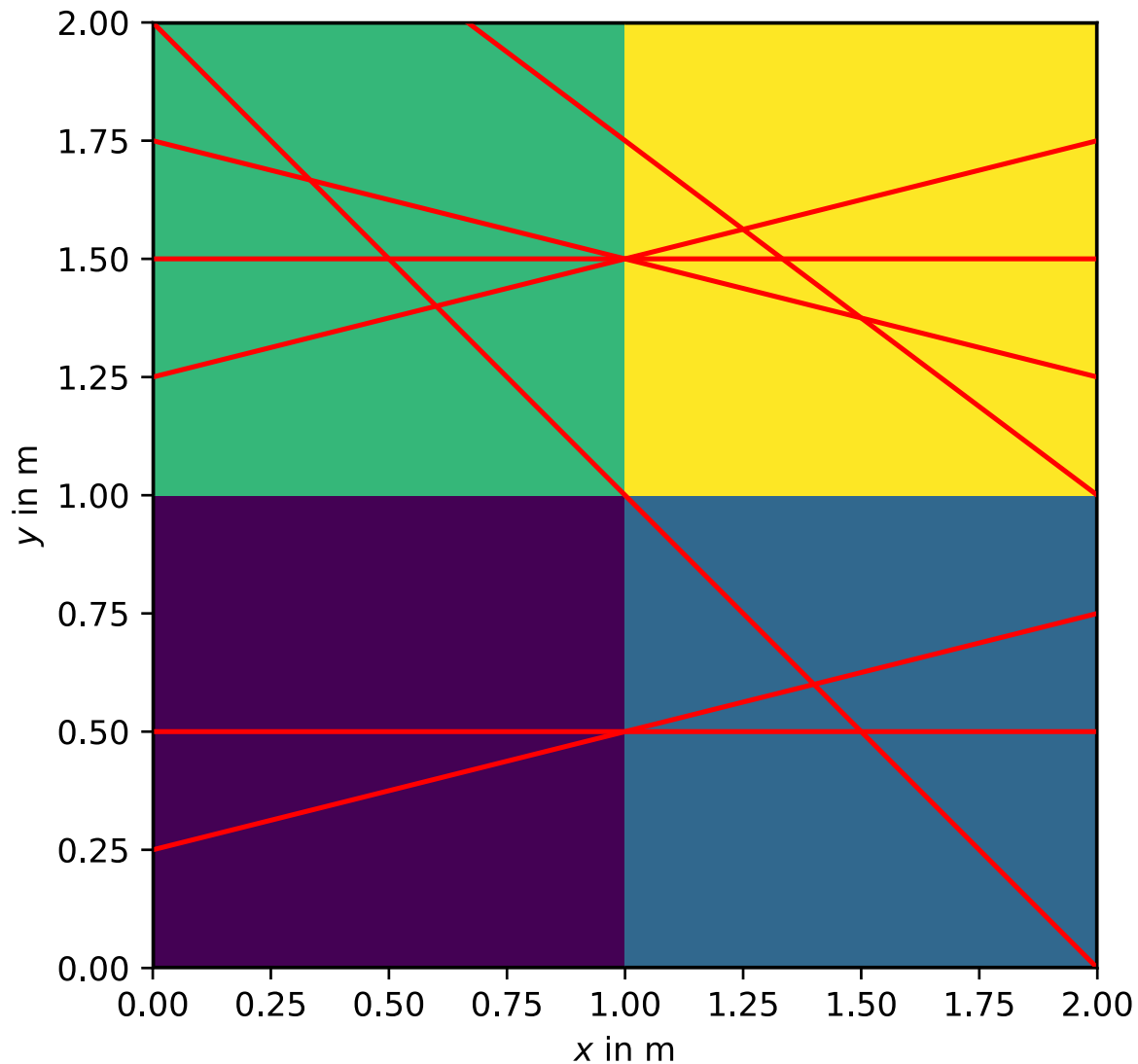
- guess whether it is a) over-, b) under-, or c) mixed-determined
- generate the ray path matrix
- determine the rank by using SVD
- generate data from an inhomogeneous model vector
- choose method & compute model
- compare with the backslash
- compile the numbers in a table

Problem types - 9



- guess whether it is a) over-, b) under-, or c) mixed-determined
- generate the ray path matrix
- determine the rank by using SVD
- generate data from an inhomogeneous model vector
- choose method & compute model
- compare with the backslash
- compile the numbers in a table

Problem types - 10



- guess a) even, b) over-, c) under-, or d) mixed-determined
- generate the ray path matrix
- determine the rank by using SVD
- generate data from an inhomogeneous model vector
- choose method & compute model
- compare with the backslash
- compile the numbers in a table

